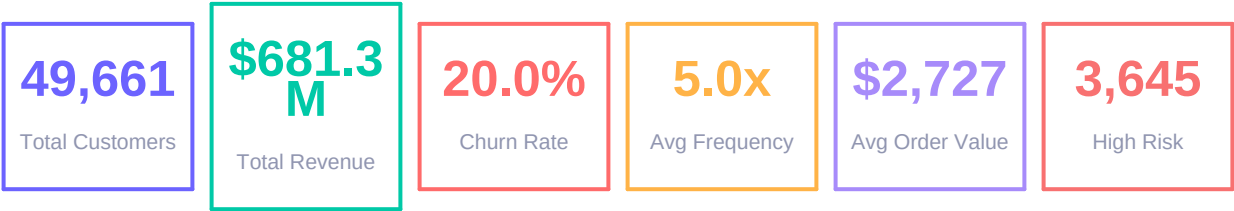


CUSTOMER ANALYTICS

Transaction & Behavior Analysis

Segmentation · Purchase Patterns · Churn Risk



E-Commerce Dataset | Jan 2020 – Sep 2023 | 500,000 Transactions

49,661 Unique Customers | 4 Product Categories | 3 Payment Methods

Executive Summary

This report presents a comprehensive analysis of e-commerce customer transactions spanning January 2020 to September 2023. Using RFM (Recency, Frequency, Monetary) scoring and K-Means clustering, 49,661 customers were segmented into four behavioral groups. A composite churn risk model identifies customers most likely to disengage, enabling targeted retention strategies across segments.

Methodology

- **Data:** Two datasets merged — 500,000 transactions, 13 features, Jan 2020–Sep 2023.
- **Cleaning:** Date parsing, null imputation, duplicate column removal, feature engineering.
- **RFM Analysis:** Recency, Frequency, and Monetary scored 1–5 per customer via quintile binning.
- **Segmentation:** K-Means (k=4) validated with Elbow Method and Silhouette Score.
- **Churn Risk:** Composite 0–1 score — Recency 40%, Frequency 30%, Monetary 20%, Return Rate 10%.

Key Findings at a Glance

Finding	Value	Implication
Largest revenue driver	Loyal Customers (43%)	Protect with retention programs
Highest individual LTV	Champions (\$24,491 avg)	VIP treatment delivers ROI
High churn risk customers	3,645 (7.3%)	Immediate re-engagement needed
Hibernating high-LTV	7,588 customers	Prime win-back opportunity
Overall churn rate	~20% across segments	Systemic experience issue
Category revenue spread	Even across 4 categories	No single-category dependency

1. Customer Segmentation

K-Means clustering on standardised RFM features produced four distinct customer segments. Each represents a meaningfully different behavioral profile requiring a tailored engagement strategy.

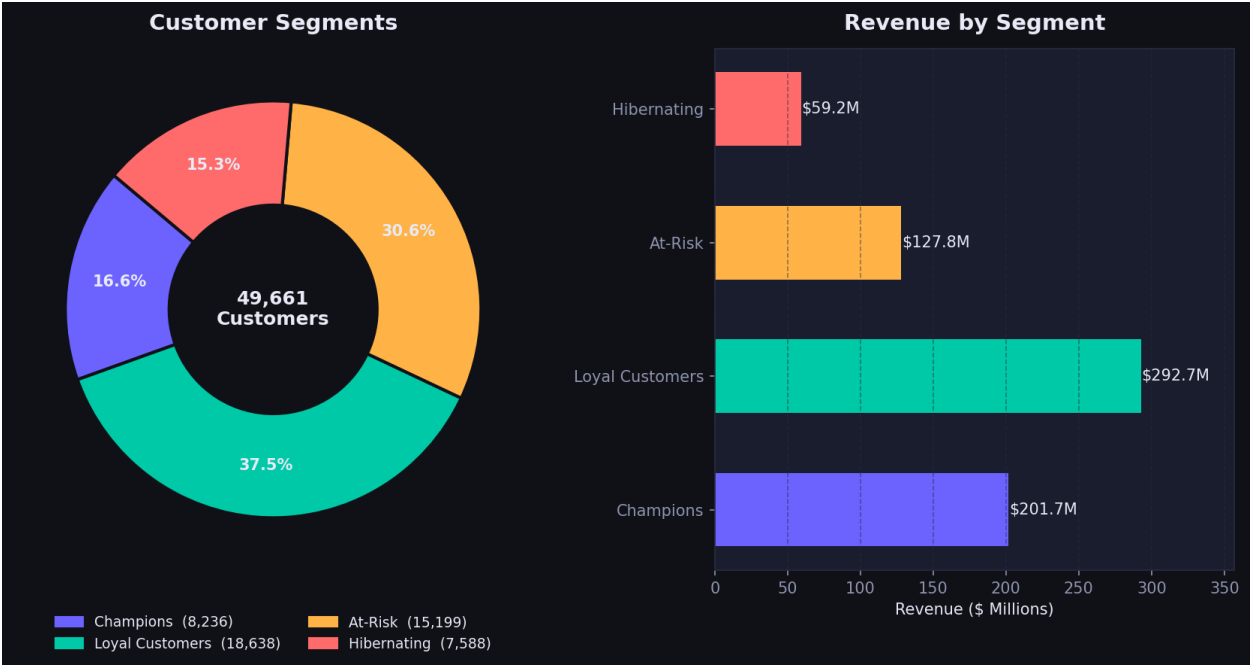


Figure 1 — Segment distribution (left) and revenue contribution by segment (right)

Champions8,236 customers 29.6% of revenue				
Avg LTV	Avg Frequency	Avg Order Value	Churn Rate	Avg Recency
\$24,491	8.5x	\$2,929	20%	140d

Loyal Customers18,638 customers 43.0% of revenue				
Avg LTV	Avg Frequency	Avg Order Value	Churn Rate	Avg Recency
\$15,703	5.7x	\$2,793	20%	181d

At-Risk15,199 customers 18.8% of revenue				
Avg LTV	Avg Frequency	Avg Order Value	Churn Rate	Avg Recency
\$8,407	3.4x	\$2,554	20%	198d

Hibernating				7,588 customers 8.7% of revenue	
Avg LTV	Avg Frequency	Avg Order Value	Churn Rate	Avg Recency	
\$7,802	2.9x	\$2,692	20%	723d	

2. Purchase Pattern Analysis

Transaction data was analysed across time, product category, and customer demographics to surface actionable patterns in buying behaviour.

2.1 Revenue Trend Over Time

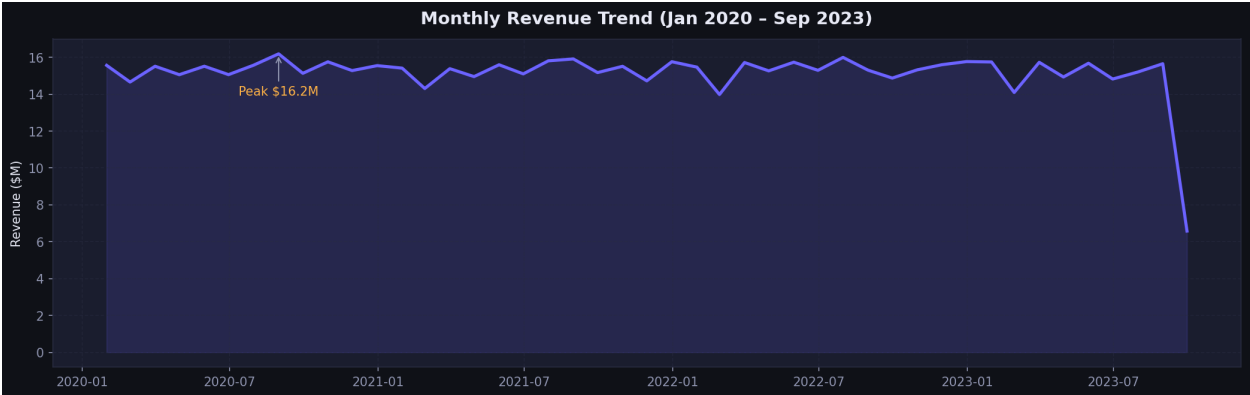


Figure 2 — Monthly revenue Jan 2020–Sep 2023

Revenue shows consistent month-to-month stability with no pronounced seasonal spikes, indicating a broad customer base not reliant on promotional events. The peak month reached \$16.2M. A slight softening in late 2023 warrants monitoring and proactive retention action.

2.2 Category & Churn Risk Breakdown

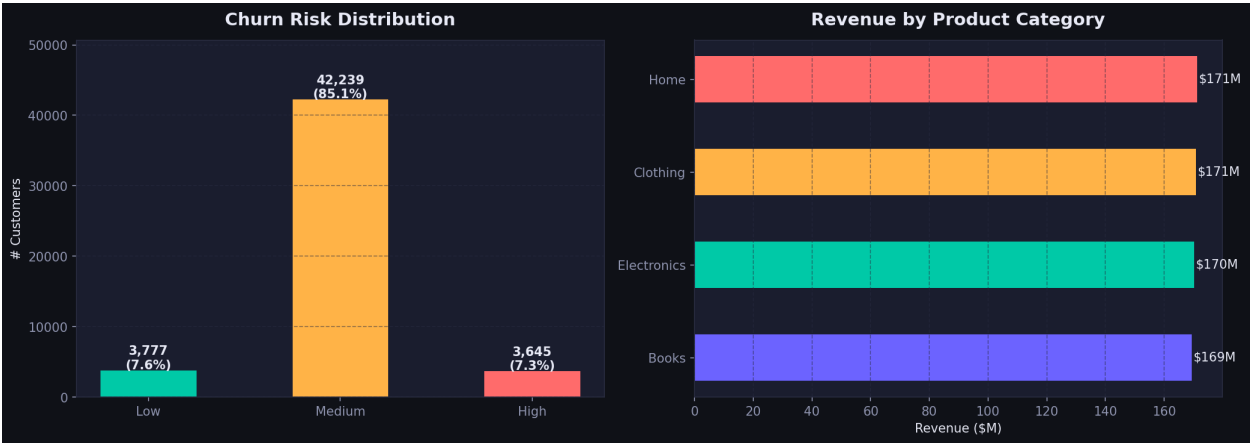


Figure 3 — Churn risk distribution (left) and revenue by product category (right)

Category Insights

Revenue is distributed nearly equally across Electronics, Clothing, Home, and Books — each contributing roughly \$170M. This diversification reduces risk but suggests limited cross-sell depth within customer journeys.

Churn Risk Insights

85% of customers are Medium risk. The 3,645 High-risk customers represent disproportionate lost revenue potential. Low-risk customers (7.6%) are stable advocates who can anchor referral programs.

3. Churn Risk Analysis

A composite churn risk score (0–1) was computed for each customer using four weighted behavioural signals. Customers labelled High Risk are candidates for immediate re-engagement.

3.1 Risk Score Methodology

Signal	Weight	Rationale
Recency (days since last purchase)	40%	Strongest predictor — long inactivity signals disengagement
Inverse Frequency	30%	Infrequent buyers are less loyal and easier to lose
Inverse Monetary Value	20%	Low spenders have weaker brand affinity
Return Rate	10%	High returns indicate product or experience dissatisfaction

3.2 Lifetime Value vs Recency by Segment

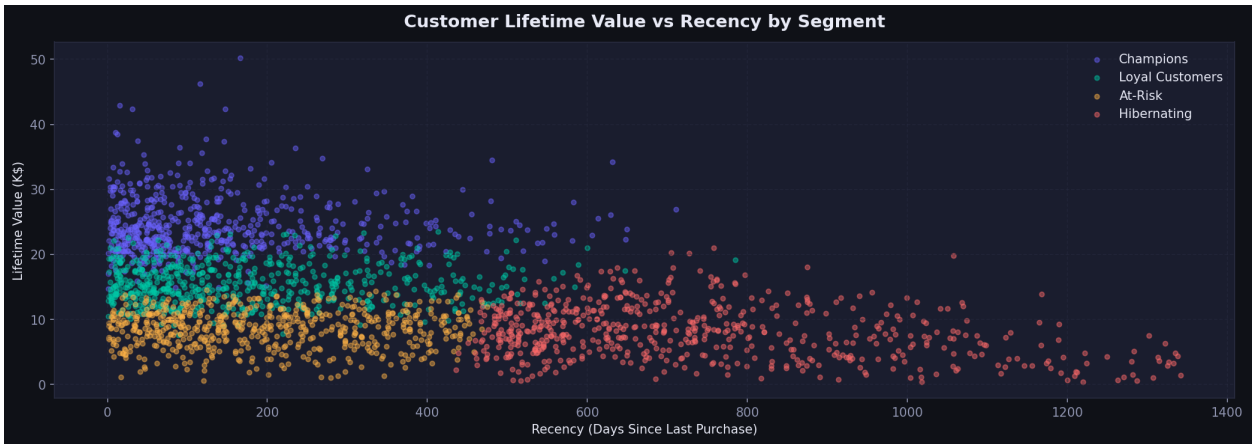


Figure 4 — Hibernating customers (red) cluster top-right: high recency, often high LTV — prime win-back targets

3.3 Top High-Risk Customers by Lifetime Value

These customers are High Risk sorted by LTV — highest revenue at immediate risk of churn.

Customer ID	Segment	Days Inactive	Purchases	Lifetime Value	Risk Score
48476	Hibernating	1058	5	\$19,807	0.727
14548	Hibernating	1078	5	\$19,522	0.714
2226	Hibernating	999	5	\$19,211	0.691
28362	Hibernating	986	5	\$18,384	0.671
28025	Hibernating	884	4	\$17,637	0.671

23120	Hibernating	936	4	\$17,376	0.688
30476	Hibernating	812	5	\$16,918	0.665
45340	Hibernating	1022	4	\$16,905	0.740
17220	Hibernating	1007	5	\$16,790	0.683
26160	Hibernating	989	5	\$16,701	0.678

4. Recommendations

Each customer segment requires a distinct engagement strategy. Recommendations are prioritised by revenue impact and implementation urgency.

Champions

- Launch a tiered VIP loyalty program with exclusive perks, early product access, and dedicated support.
- Use Champions as brand ambassadors — referral programs drive high-trust, low-cost acquisition.
- Offer co-creation opportunities: beta testing, product feedback panels, or community roles.
- Maintain momentum with surprise-and-delight moments — personalised gifts, priority service.

Loyal Customers

- Protect this segment aggressively — they generate 43% of revenue and are your most reliable base.
- Deploy personalised product recommendations built on past category purchase behaviour.
- Introduce member-only discount tiers to reinforce loyalty and deter competitive switching.
- Upsell into higher-margin product lines using purchase history signals.

At-Risk

- Trigger automated win-back sequences: "We miss you" emails with time-limited discount codes.
- Run short NPS or exit-intent surveys to diagnose the root cause of disengagement.
- Offer free shipping, bundle deals, or category-specific promotions to reduce switching friction.
- Escalate high-LTV At-Risk customers to human outreach — account managers or live chat.

Hibernating

- Prioritise re-engagement by LTV — focus on customers with lifetime value above \$15,000 first.
- Use aggressive reactivation offers: 25–40% discount bundles or free-gift-with-purchase.
- Test SMS and push notification channels where email open rates are low for dormant segments.
- Set a 90-day win-back deadline — customers inactive 900+ days have very low reactivation odds.

Priority action: Load `high_churn_risk_customers.csv` into your CRM immediately. These 3,645 customers represent the highest revenue-at-risk and should enter a re-engagement workflow within 48 hours.

Appendix — Full Segment Profile

Complete statistical summary across all four customer segments.

Metric	Champions	Loyal Customers	At-Risk	Hibernating
Customers	8,236	18,638	15,199	7,588
Revenue Share	29.6%	43.0%	18.8%	8.7%
Avg LTV	\$24,491	\$15,703	\$8,407	\$7,802
Avg Order Value	\$2,929	\$2,793	\$2,554	\$2,692
Avg Frequency	8.5x	5.7x	3.4x	2.9x
Avg Recency (days)	140	181	198	723
Return Rate	0.40	0.41	0.41	0.40
Churn Rate	20.0%	20.0%	20.0%	20.0%

Source: ecommerce_customer_data_large.csv + ecommerce_customer_data_custom_ratios.csv | 500,000 transactions | 49,661 unique customers | Jan 2020–Sep 2023

Methodology: RFM Analysis + K-Means Clustering (k=4) + Composite Churn Risk Scoring | Tools: Python, pandas, scikit-learn, matplotlib, ReportLab